```
# Task1
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_marks():
    Name = name.get()
    Python = int(python.get())
    Java = int(java.get())
    C = int(c.get())
    SQL = int(sql.get())

    data = {
        "Subject":["Python","Java","C","SQL"],
        "Marks":[Python,Java,C,SQL]
        }
    df = pd.DataFrame(data)

    total = np.sum(df["Marks"])
    average = np.mean(df["Marks"])
    highest = np.max(df["Marks"])
    lowest = np.min(df["Marks"])

    result.config(text=f"Total Marks: {total}\n Average Marks: {average}\n Highest Marks: {highest}\n
Lowest Marks: {lowest}")

    plt.bar(df["Subject"],df["Marks"])
    plt.title(f"{Name} - Subject wise Marks")
    plt.xlabel("Subjects")
    plt.ylabel("Marks")
    plt.show()


root = tk.Tk()
root.title("Student Result Analysis System")
```

```
root.geometry("500x500")
root.configure(bg="white")

heading = tk.Label(root, text="STUDENT RESULT ANALYSIS SYSTEM", font=("Calibri",20,"bold"),
                fg = "white",bg="blue")
heading.pack()

tk.Label(root,text = "Student Name").pack()
name = tk.Entry(root)
name.pack()

tk.Label(root,text = "Python Marks").pack()
python = tk.Entry(root)
python.pack()

tk.Label(root,text = "Java Marks").pack()
java = tk.Entry(root)
java.pack()

tk.Label(root,text = "C Marks").pack()
c = tk.Entry(root)
c.pack()

tk.Label(root,text = "SQL Marks").pack()
sql = tk.Entry(root)
sql.pack()

tk.Button(root,text = "Analyze Result",command=analyze_marks).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

```
# Task2
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_expense():
    food_amt = float(food.get())
    travel_amt = float(travel.get())
    shopping_amt = float(shopping.get())
    other_amt = float(other.get())

    data = {
        "Category":["Food","Travel","Shopping","Other"],
        "Amount":[food_amt,travel_amt,shopping_amt,other_amt]
    }

    df = pd.DataFrame(data)

    total = np.sum(df["Amount"])
    average = np.mean(df["Amount"])

    result.config(text=f"Total Expense: {total}\nAverage Expense: {average}")

    plt.pie(df["Amount"],labels=df["Category"],autopct='%1.1f%%')
    plt.title("Expense Distribution")
    plt.show()

root = tk.Tk()
root.title("Personal Expense Tracker")
root.geometry("500x500")

tk.Label(root,text="Food Expense").pack()
food = tk.Entry(root)
food.pack()
```

```
tk.Label(root,text="Travel Expense").pack()
travel = tk.Entry(root)
travel.pack()

tk.Label(root,text="Shopping Expense").pack()
shopping = tk.Entry(root)
shopping.pack()

tk.Label(root,text="Other Expense").pack()
other = tk.Entry(root)
other.pack()

tk.Button(root,text="Analyze Expense",command=analyze_expense).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

 # Task3

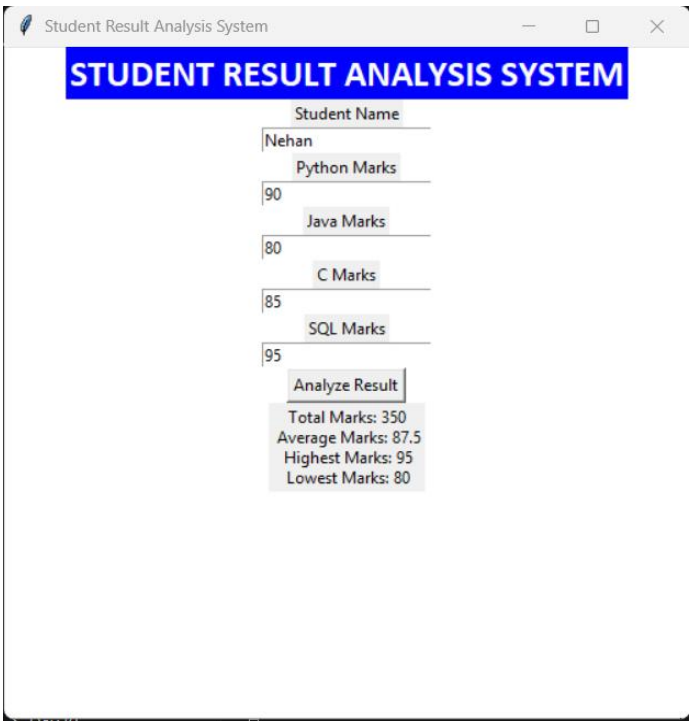
```
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_books():
    book1_copies = int(book1.get())
    book2_copies = int(book2.get())
    book3_copies = int(book3.get())

    data = {
        "Book":["Python","Java","C++"],
        "Copies":[book1_copies,book2_copies,book3_copies]
    }
```

```
    df = pd.DataFrame(data)

    total = np.sum(df["Copies"])
    average = np.mean(df["Copies"])

    result.config(text=f"Total Books: {total}\nAverage Copies: {average}")

    plt.bar(df["Book"],df["Copies"])
    plt.title("Library Book Distribution")
    plt.xlabel("Books")
    plt.ylabel("Copies")
    plt.show()

root = tk.Tk()
root.title("Library Book Data Analyzer")
root.geometry("500x500")

tk.Label(root,text="Python Book Copies").pack()
book1 = tk.Entry(root)
book1.pack()

tk.Label(root,text="Java Book Copies").pack()
book2 = tk.Entry(root)
book2.pack()

tk.Label(root,text="C++ Book Copies").pack()
book3 = tk.Entry(root)
book3.pack()

tk.Button(root,text="Analyze Books",command=analyze_books).pack()

result = tk.Label(root,text="")
result.pack()
root.mainloop()
```

```
# Task4
import tkinter as tk
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

def analyze_quiz():
    s1 = int(student1.get())
    s2 = int(student2.get())
    s3 = int(student3.get())
    s4 = int(student4.get())

    data = {
        "Student":["A","B","C","D"],
        "Score":[s1,s2,s3,s4]
    }

    df = pd.DataFrame(data)

    average = np.mean(df["Score"])
    highest = np.max(df["Score"])

    result.config(text=f"Average Score: {average}\nHighest Score: {highest}")

    plt.bar(df["Student"],df["Score"])
    plt.title("Quiz Results")
    plt.xlabel("Students")
    plt.ylabel("Score")
    plt.show()

root = tk.Tk()
root.title("Quiz Result Analyzer")
root.geometry("500x500")

tk.Label(root,text="Student A Score").pack()
```

```
student1 = tk.Entry(root)
student1.pack()

tk.Label(root,text="Student B Score").pack()
student2 = tk.Entry(root)
student2.pack()

tk.Label(root,text="Student C Score").pack()
student3 = tk.Entry(root)
student3.pack()

tk.Label(root,text="Student D Score").pack()
student4 = tk.Entry(root)
student4.pack()

tk.Button(root,text="Analyze Quiz",command=analyze_quiz).pack()

result = tk.Label(root,text="")
result.pack()

root.mainloop()
```

<u>Outputs:</u>



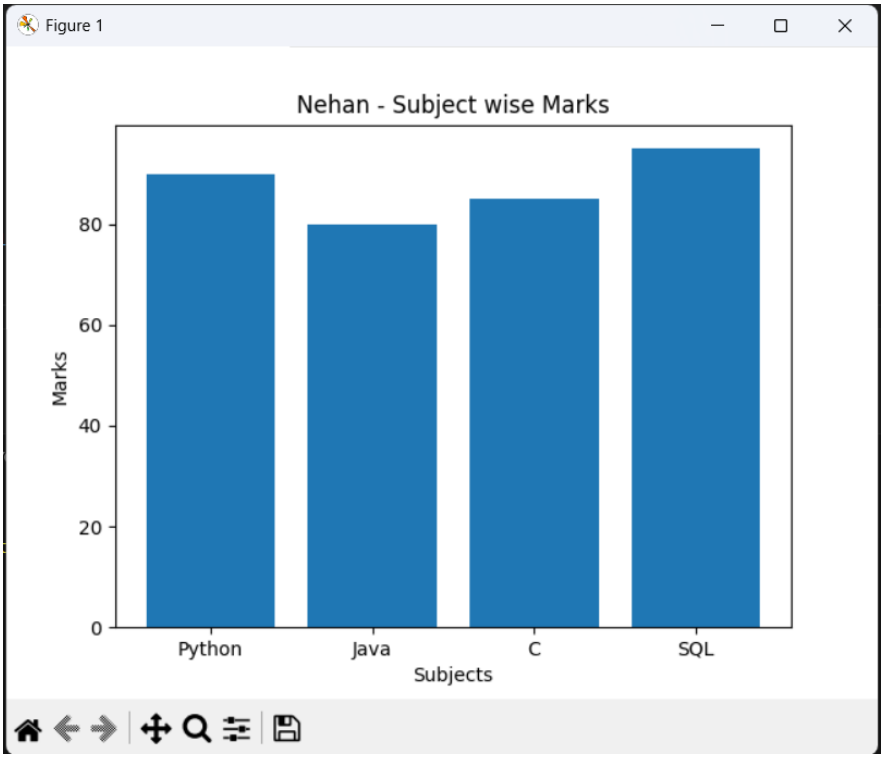
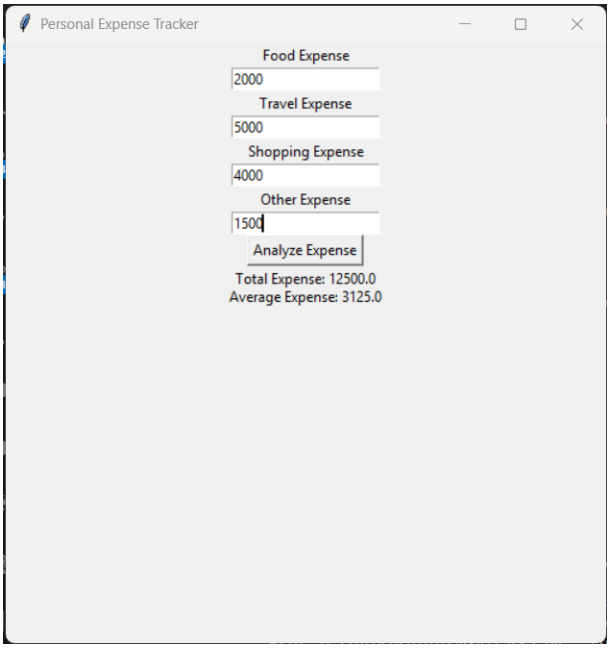


Figure 1
Nehan - Subject wise Marks
Marks
0
20
40
60
80
Python
Java
C
SQL
Subjects


Personal Expense Tracker
Food Expense
2000
Travel Expense
5000
Shopping Expense
4000
Other Expense
1500
Analyze Expense
Total Expense: 12500.0
Average Expense: 3125.0

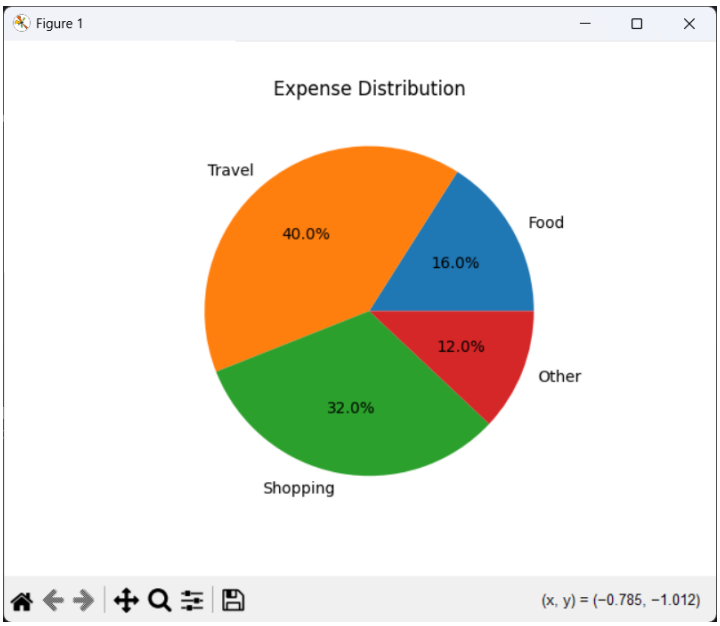
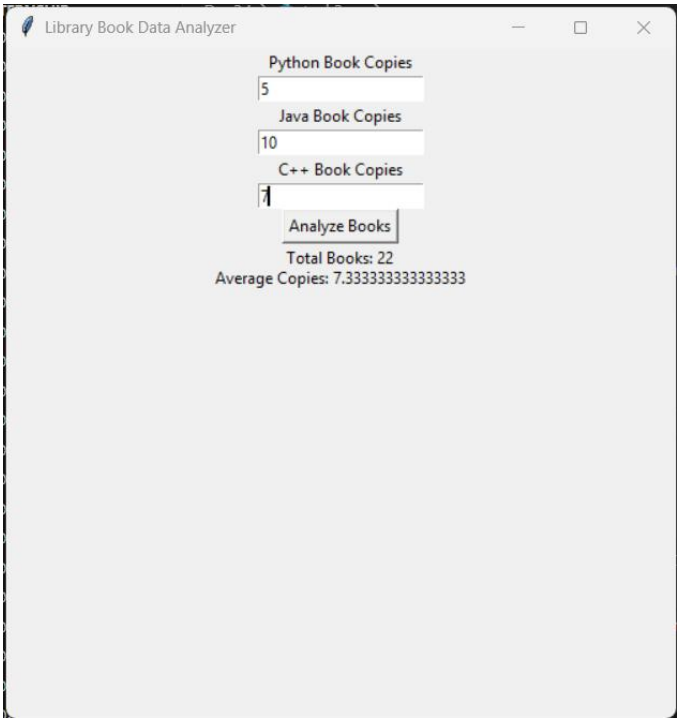


Figure 1
Expense Distribution
Travel
40.0%
Food
16.0%
12.0%
Other
32.0%
Shopping
(x, y) = (−0.785, −1.012)


Library Book Data Analyzer
Python Book Copies
5
Java Book Copies
10
C++ Book Copies
7
Analyze Books
Total Books: 22
Average Copies: 7.333333333333333

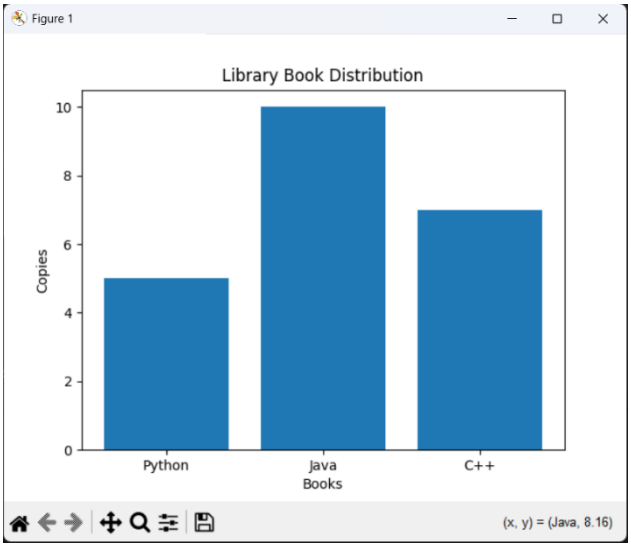
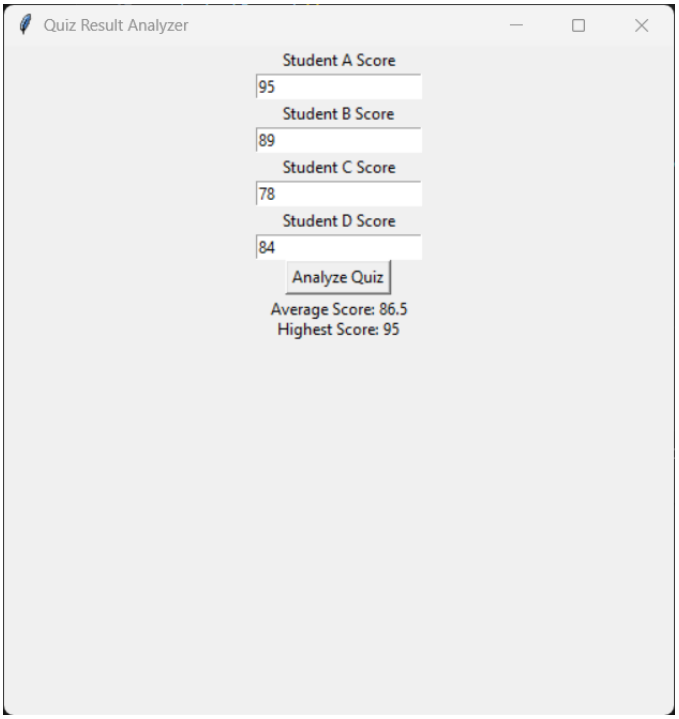


Figure 1
Library Book Distribution
Copies
0
2
4
6
8
10
Python
Java
C++
Books
(x, y) = (Java, 8.16)


Quiz Result Analyzer
Student A Score
95
Student B Score
89
Student C Score
78
Student D Score
84
Analyze Quiz
Average Score: 86.5
Highest Score: 95

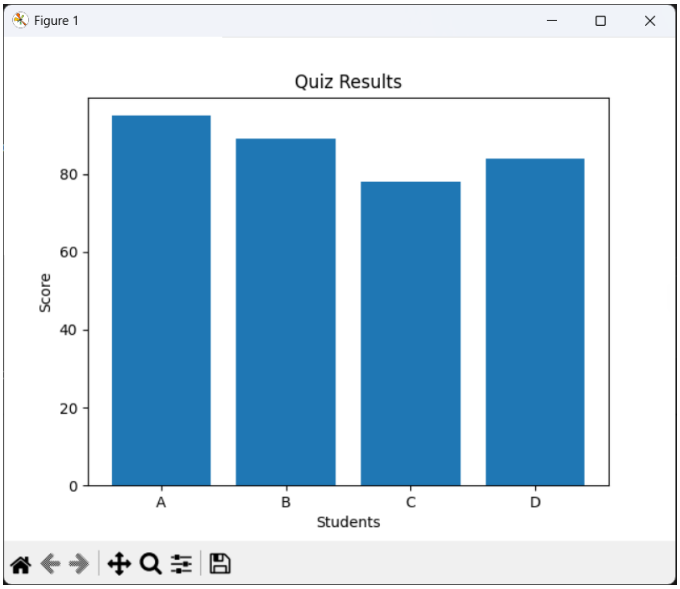


Figure 1
Quiz Results
Score
0
20
40
60
80
A
B
C
D
Students